Data correction/standardisation [Week 3]

- Most data is not perfect and requires some level of correction to a standardised reference (terrain category 2 (open terrain), flat, 10 m height, 3-second gust (0.2-second used in AS/NZS1170.2), 10-minute mean [WMO]).
- Typical corrections:
    - Instrumentation
    - Gust duration
    - Height
    - Topography/terrain
    - Station movement

CIVL4340
Wind Engineering

Week 4: Atmospheric boundary layer flows

The University of Queensland
Australia

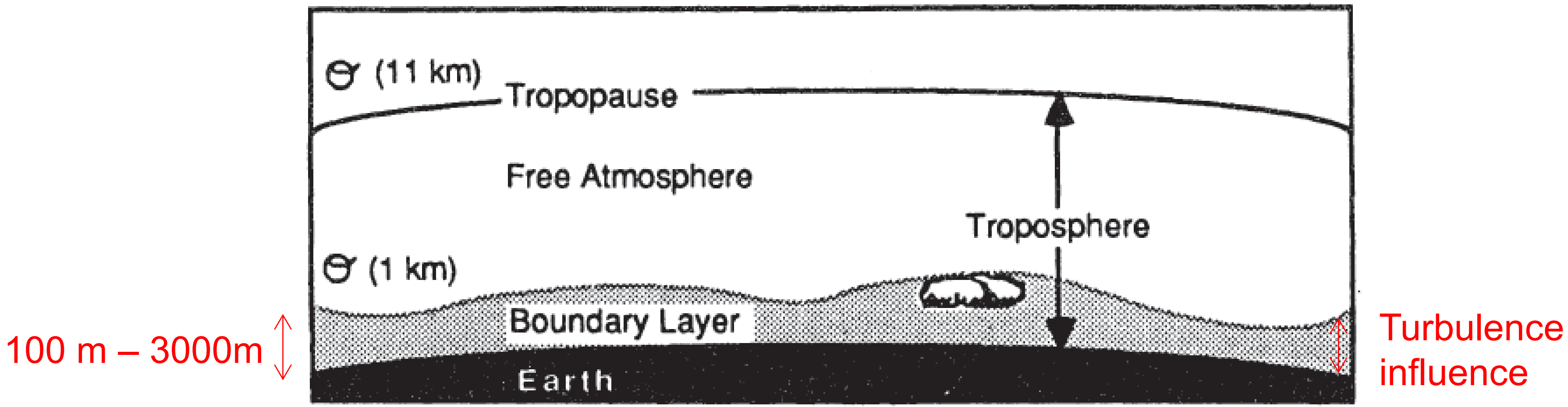
Overview

- Boundary layer theory and concepts
- The turbulent wind

Reading

- Holmes, Chapter 2 & 3
- Simiu and Yeo, Chapter 2 & 3

Boundary layer concepts



(Stull 1989)

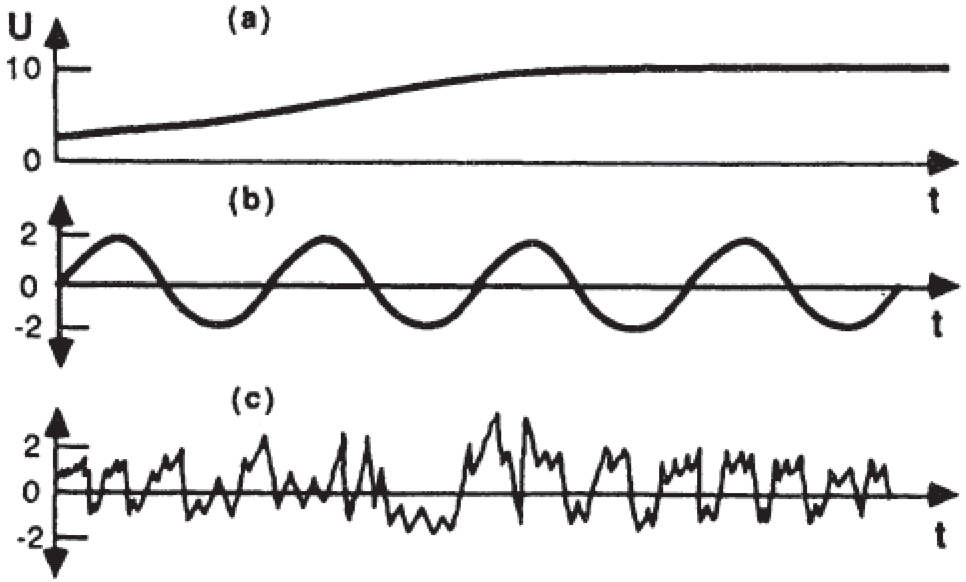
- *"The boundary layer is that part of the troposphere that is directly influenced by the presence of the earth's surface, and responds to surface forcings within a timescale of about an hour or less.* (Stull)
- Forcings: friction drag, evaporation, transpiration, heat transfer, pollutant emissions, terrain induced flow.
- Boundary layer depth varies with type of wind event (e.g. shallower for thunderstorm outflows than synoptic gales).

Boundary layer concepts

$\bar{U}$ Mean wind

Waves

$u(t)$ Turbulence



Govern horizontal movements

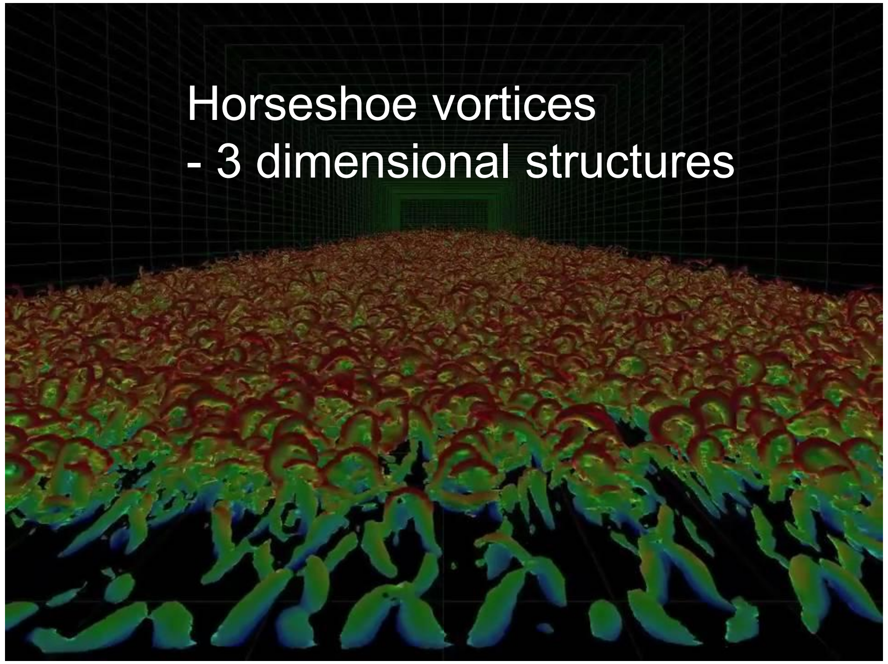
Govern vertical movements

- In wind engineering we typically ignore waves.
- We divide wind into mean and fluctuating wind speed components.

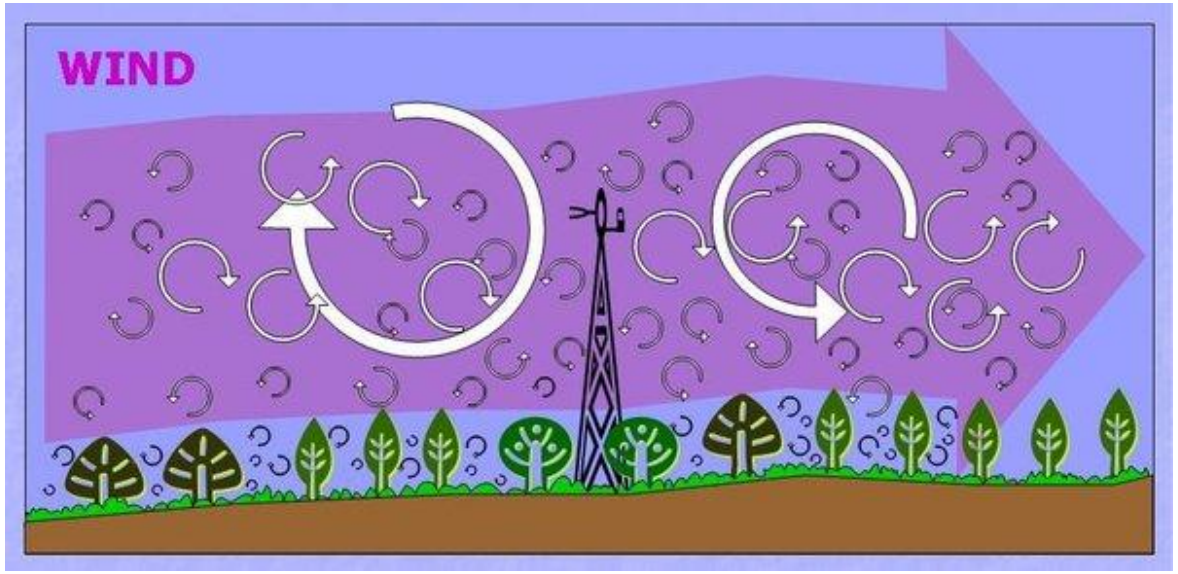
$$U(t) = \bar{U} + u(t)$$

Boundary layer concepts

- Turbulence is vitally important!
- The boundary layer responds to forcings through turbulent transport. Diffusion does this poorly (hence the free atmosphere doesn't "see" the ground).
- Turbulent exists across a range of spatial and temporal scales.



https://www.youtube.com/watch?v=Wr984EOmNaY

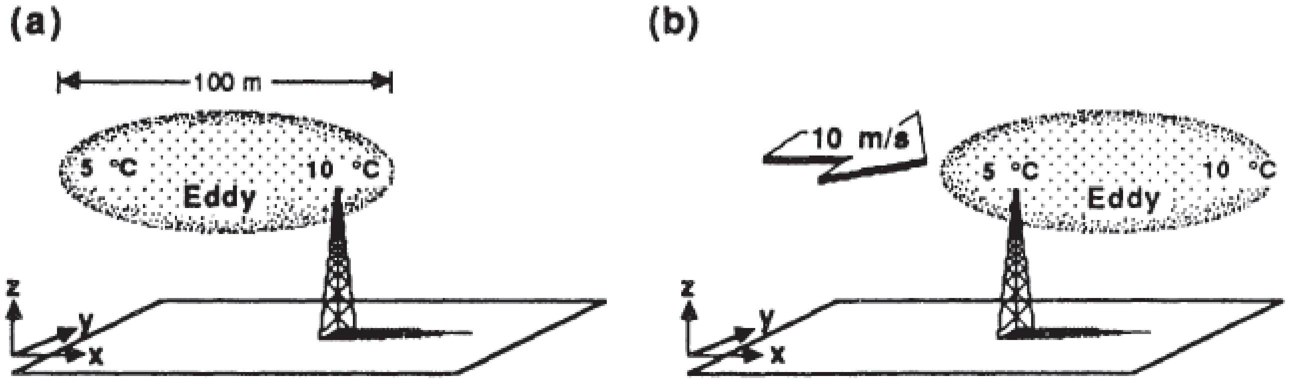


http://upload.wikimedia.org/wikipedia/en/1/11/EddyCovariance_diagram_1.jpg

https://www.youtube.com/watch?v=e1TbkLIDWys

Boundary layer concepts

Taylor's frozen turbulence



(Stull 1989)

$$\frac{\partial \xi}{\partial t} = -U\frac{\partial \xi}{\partial x} - V\frac{\partial \xi}{\partial y} - W\frac{\partial \xi}{\partial z}$$

Concept used by wind engineers to understand what size eddies will influence different scales of structures.

Governing equations (Navier-Stokes)

- Continuity equation [C], Equations of motion [M].
- In wind engineering we typically ignore energy equation [E].
- Averaging [C] and [M] with time, we get:

$$U\frac{\partial U}{\partial x}+V\frac{\partial U}{\partial y}+W\frac{\partial U}{\partial z}+\frac{1}{\rho}\frac{\partial p}{\partial x}-fV-\frac{1}{\rho}\frac{\partial \tau_u}{\partial z}=0$$

$$U\frac{\partial V}{\partial x}+V\frac{\partial V}{\partial y}+W\frac{\partial V}{\partial z}+\frac{1}{\rho}\frac{\partial p}{dy}+fU-\frac{1}{\rho}\frac{\partial \tau_v}{\partial z}=0$$

$$\frac{1}{\rho}\frac{\partial p}{\partial z}+g=0$$

$$\frac{\partial U}{\partial x}+\frac{\partial V}{\partial y}+\frac{\partial W}{\partial z}=0$$

Governing equations

- Equations are 'closed' with velocity field relationships.
  - These models approximate stresses, e.g.

$$\tau_u = \rho K(x,y,z)\frac{\partial U}{\partial z}$$

Eddy viscosity

Mixing length

$$\tau_v = \rho K(x,y,z)\frac{\partial V}{\partial z}$$

$$K(x,y,z) = L^2(x,y,z)\left[\left(\frac{\partial U}{\partial z}\right)^2 + \left(\frac{\partial V}{\partial z}\right)^2\right]^{1/2}$$

- Either L or K must be defined.
- To study turbulence we must then substitute back in, $U(t) = \bar{U} + u(t)$
- This complicates things and additional 'turbulence closure' schemes are required [not covered in this course].

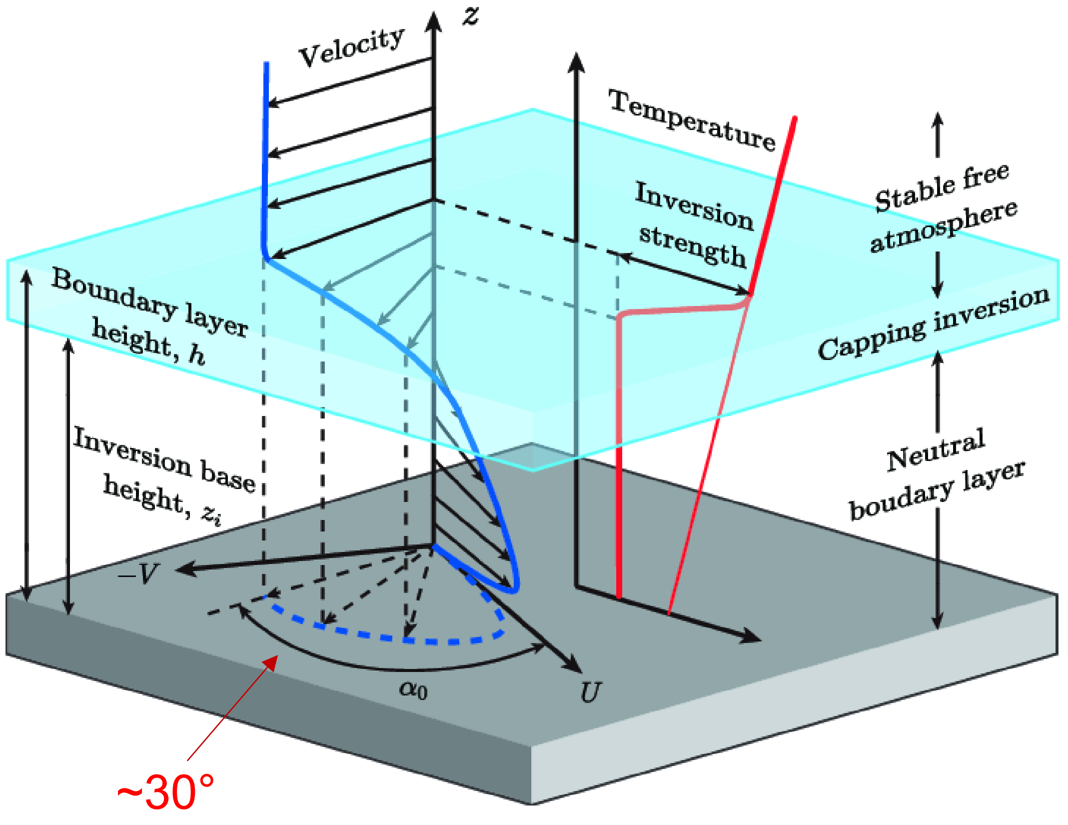
Mean velocity profiles

- Under equilibrium and horizontally homogeneous conditions,

$$V_g - V = \frac{1}{\rho f}\frac{\partial \tau_u}{\partial z} \qquad\qquad U_g - U = \frac{1}{\rho f}\frac{\partial \tau_v}{\partial z}$$

- If the stress equations from the previous slide are substituted in, we get the mean velocity profile with an Ekman spiral.

- Historically the influence of the Ekman spiral is traditionally ignored in wind engineering, but can not be for super-tall buildings, e.g. Burj Khalifa.



https://www.researchgate.net/figure/Schematic-representation-of-conventionally-neutral-ABL-not-to-scale-It-shows-the_fig1_360830733

Mean velocity profiles

- An alternate approach exists for estimating stress near the surface, based on similarity theory, where;

$$\tau_0 = F(U\mathbf{i} + V\mathbf{j}, z, z_0, \rho)$$

- Which can be rewritten as

$$\frac{U\mathbf{i} + V\mathbf{j}}{u_*} = f(z/z_0)$$

- where

$$u_* = \sqrt{\frac{\tau_0}{\rho}}$$

Log law

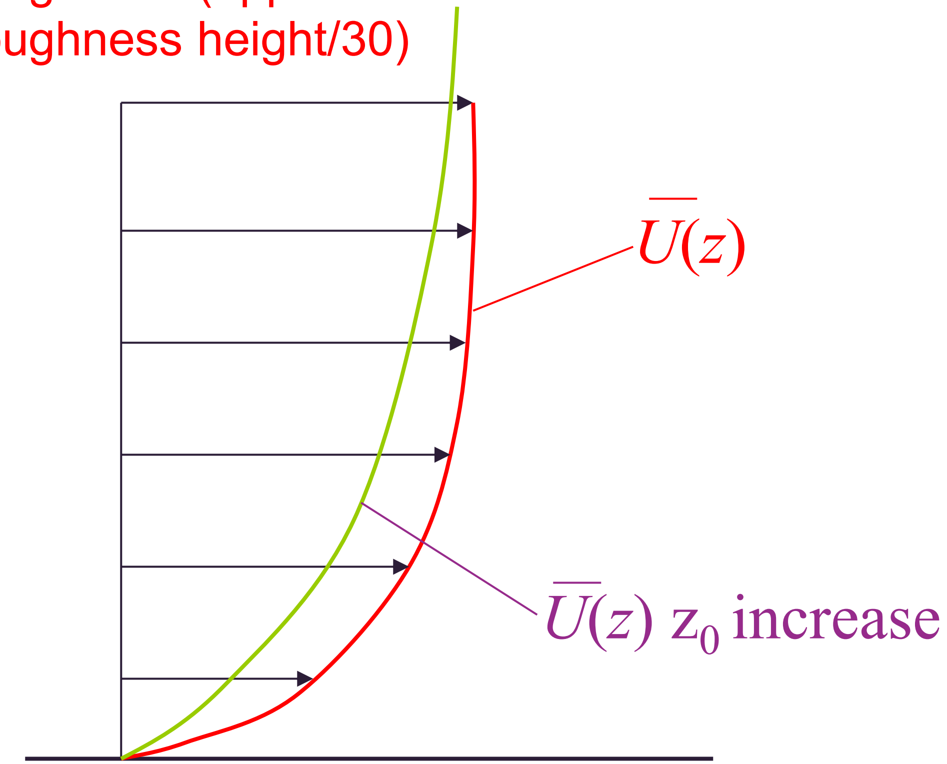
Friction velocity
(not a physical thing)

Aerodynamic
roughness (approx.
roughness height/30)

$$\overline{U}(z) = \frac{u_*}{\kappa} \ln\left(\frac{z}{z_0}\right)$$

Von Karman's
constant (usually 0.4)

$$u_* = \sqrt{\frac{\tau_0}{\rho}}$$



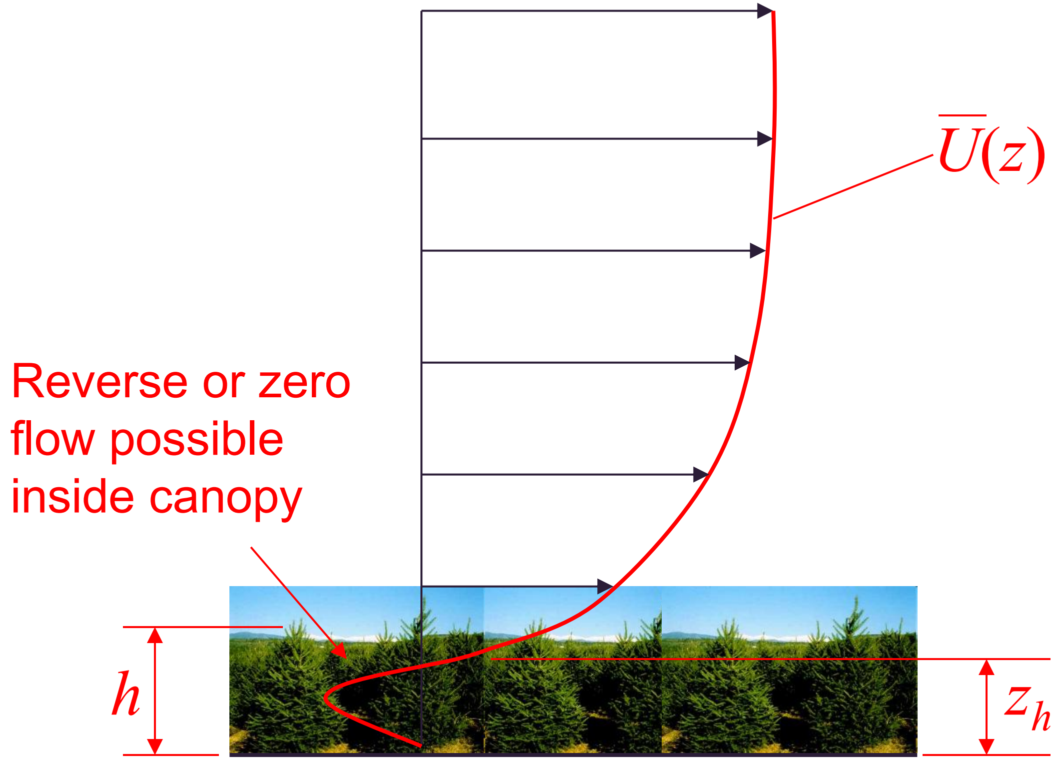
Terrain type	*Roughness length (m)*
Very flat terrain (snow, desert)	0.001–0.005
Open terrain (grassland, few trees)	0.01–0.05
Suburban terrain (buildings 3–5 m)	0.1–0.5
Dense urban (buildings 10–30 m)	1–5

Modified log law

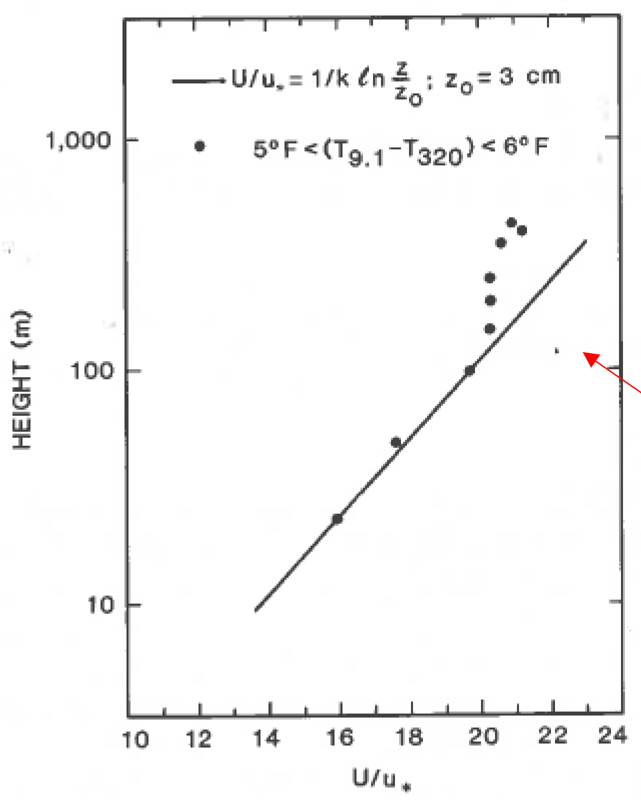
Zero plane displacement

$$\overline{U}(z) = \frac{u_*}{\kappa} \ln\left(\frac{z - z_h}{z_0}\right)$$

$$z_h \approx 0.75h$$

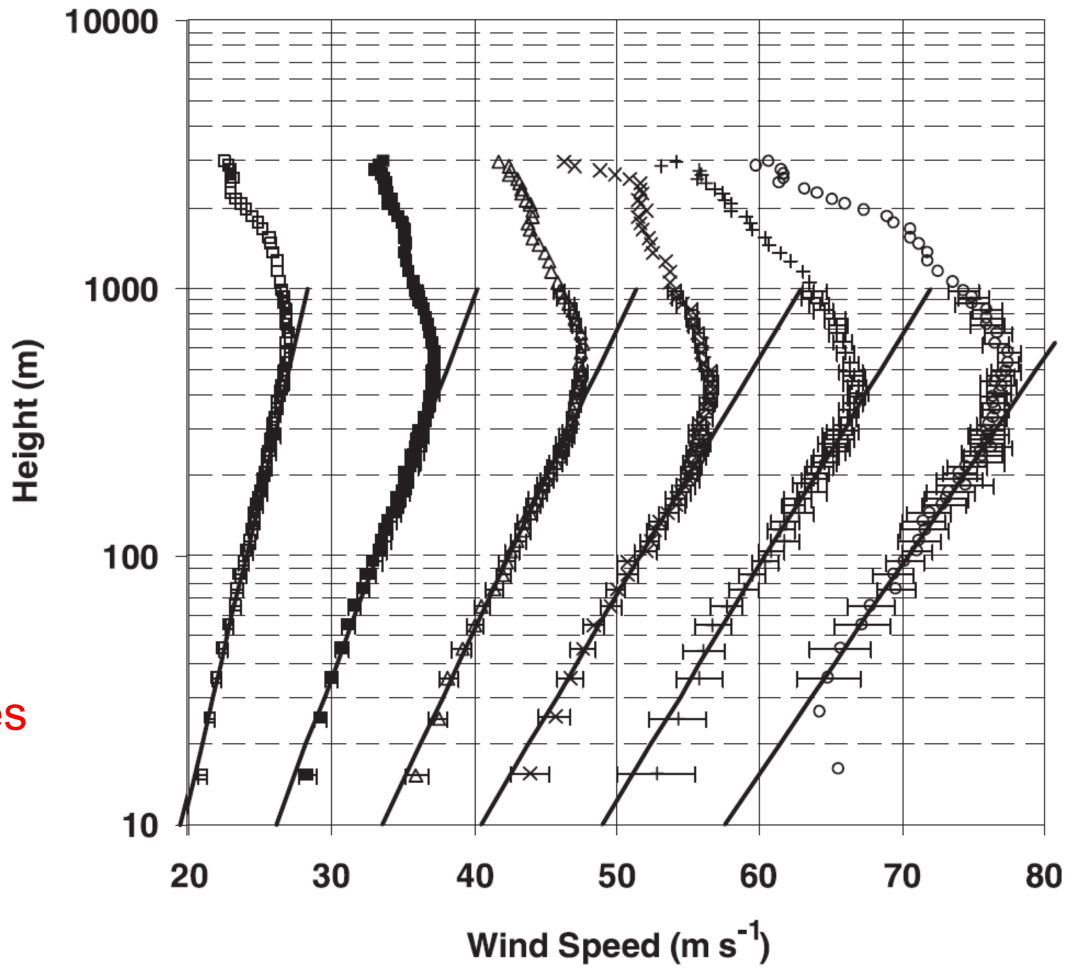


Log law



Lappe (1964)

Log law only applicable below a few hundred metres



Vickery et al. (2009)

Tropical cyclone data

Deaves and Harris Model

- Empirical equation to fit observations up to the gradient height, z_g (assumed between 500 m and 3 km)

- AS/NZS1170.2 has historically been based on this equation.

$$\overline{U}(z) = \frac{u_*}{\kappa}\left[\ln\left(\frac{z}{z_0}\right) + 5.75\left(\frac{z}{z_g}\right) - 1.88\left(\frac{z}{z_g}\right)^2 - 1.33\left(\frac{z}{z_g}\right)^3 + 0.25\left(\frac{z}{z_g}\right)^4\right]$$

$$z_g = \frac{u_*}{12\Omega \sin\lambda}$$

Gradient height

Earth rotation rate

Latitude

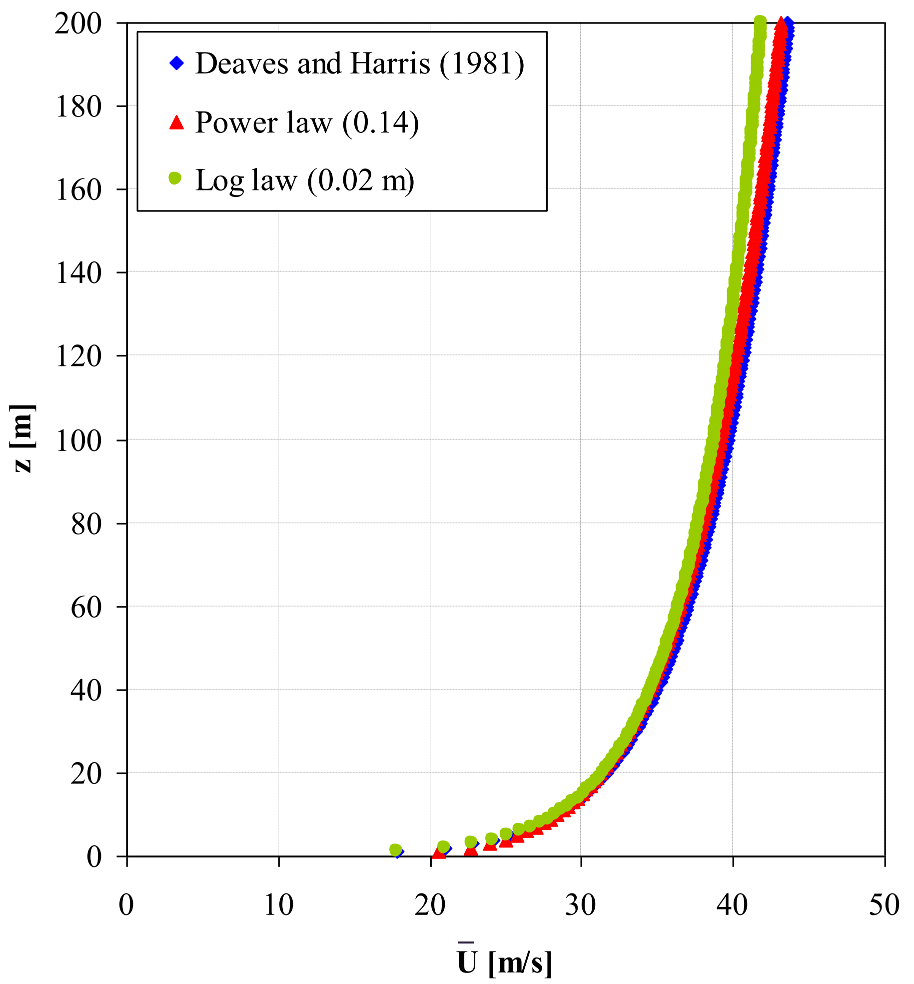
Power law

- Entirely empirical.
- Can match log law closely

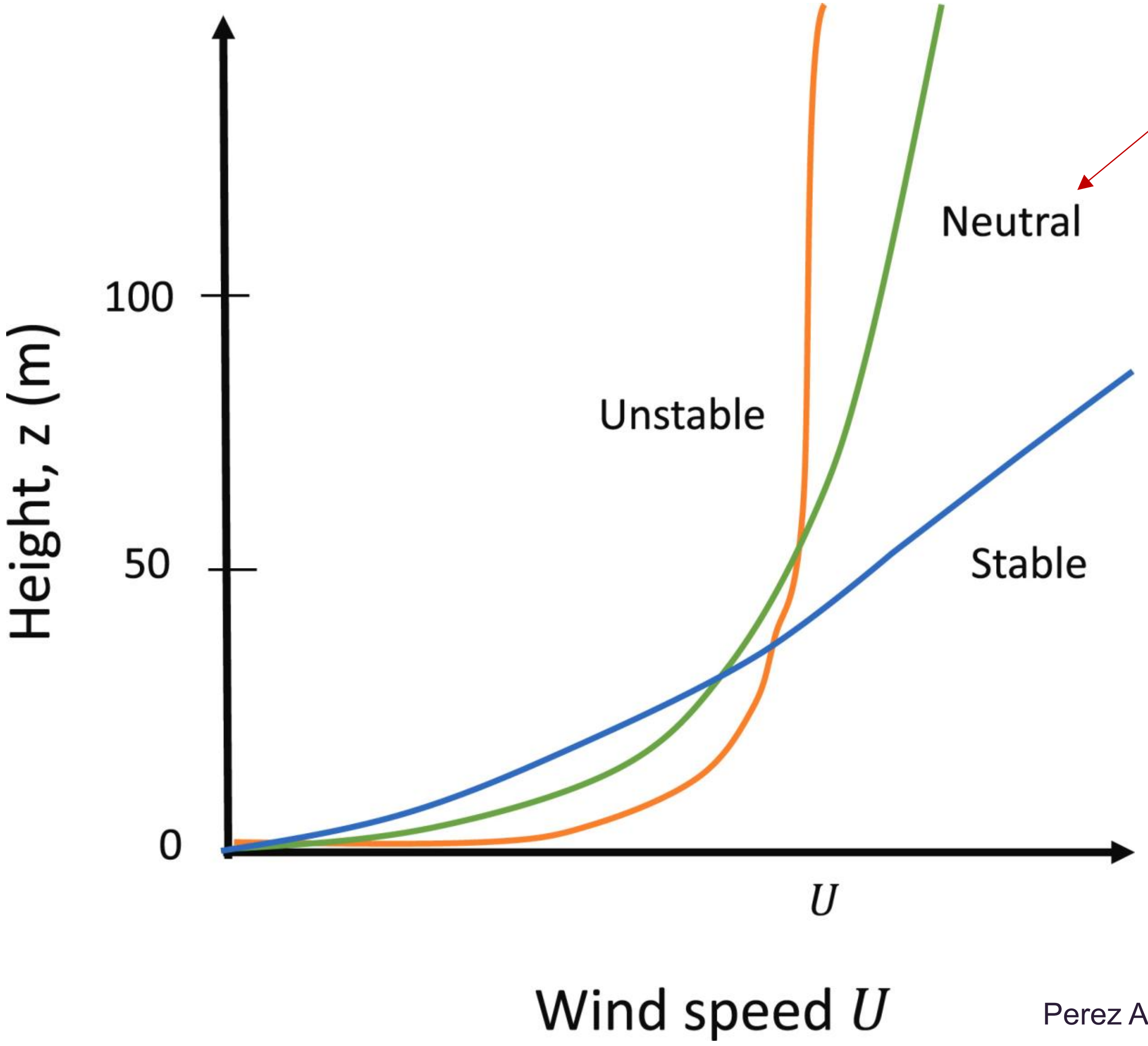
$$\overline{U}(z) = \overline{U}(\text{ref})\left(\frac{z}{z(\text{ref})}\right)^{\alpha}$$

Typically 10 m

Roughness parameter



Atmospheric stability



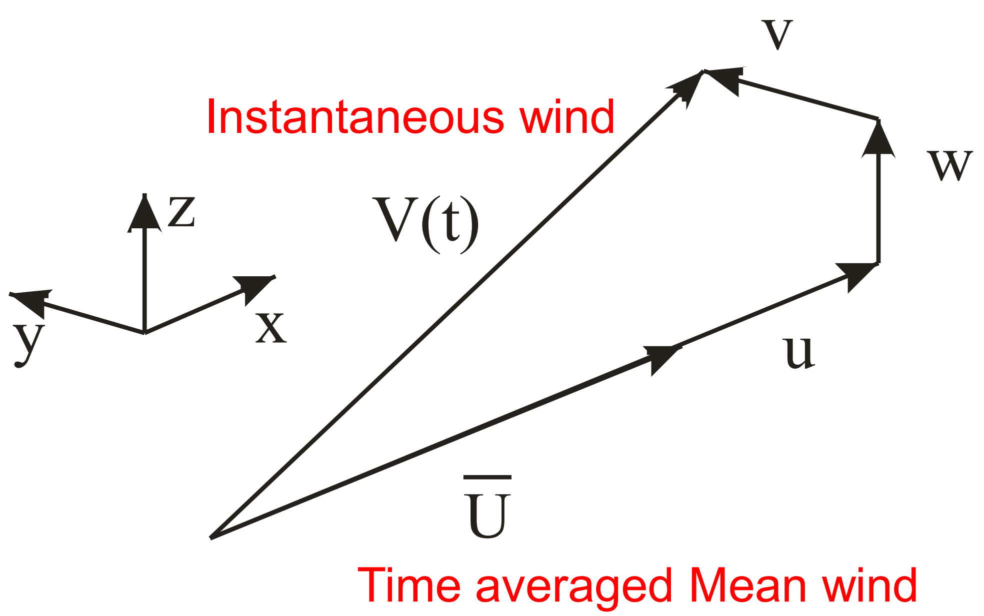
Assumed for wind loading

All important for wind energy

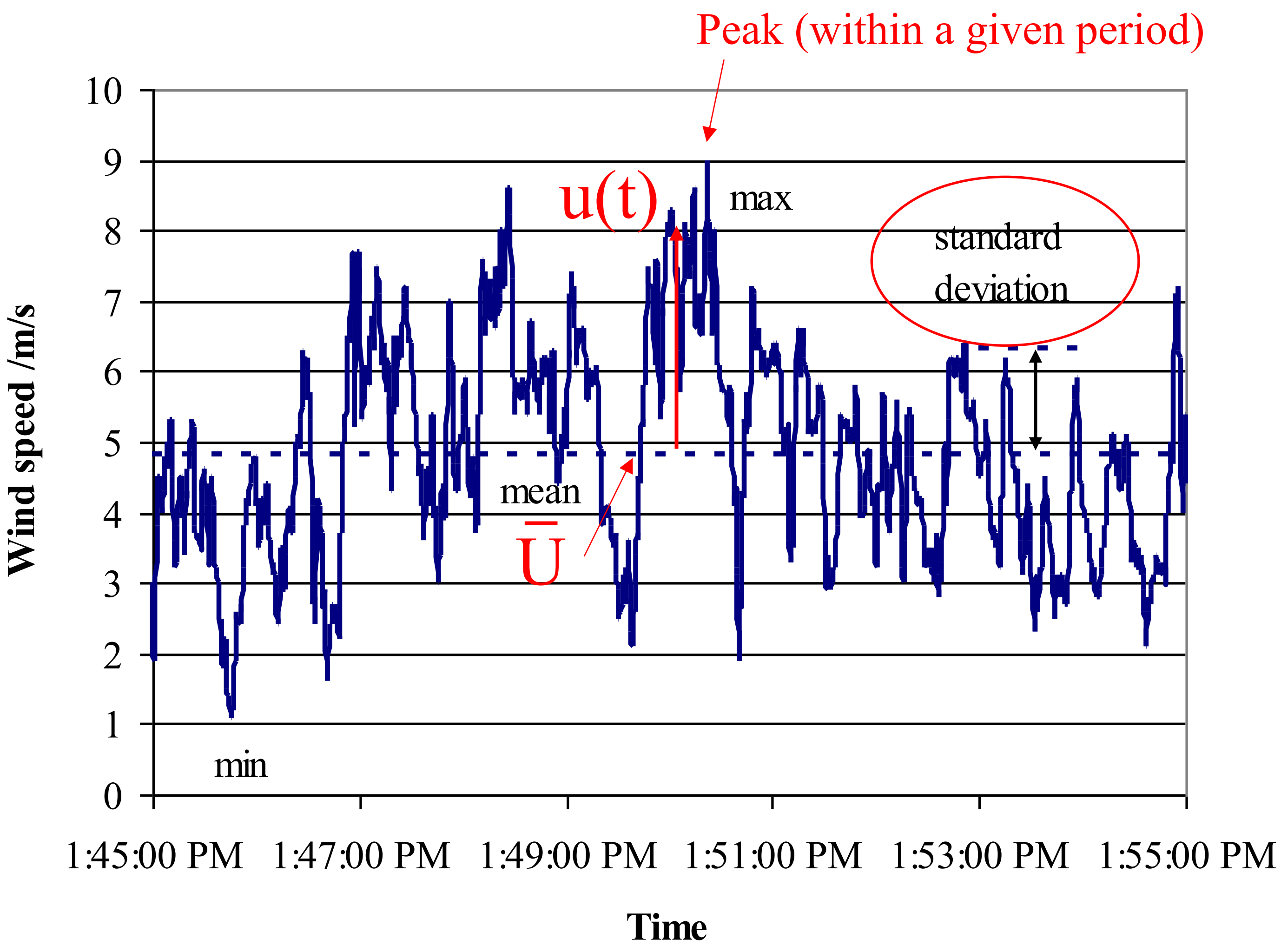
Perez Albornoz et al. (2022)

Turbulent wind

- Wind can be considered random in nature and therefore can conveniently be described using statistics.
- Wind varies in time and space.
- Dealt with by splitting into three orthogonal directions, where: $\bar{u}, \bar{v}, \bar{w} = 0$



Turbulent wind



$$V(t) = \overline{U} + u(t)$$

Mean wind

$$\overline{U} = \frac{1}{T}\int_0^T V(t)\,dt$$

Averaging period

- Standard averaging times:
  - 1 minute (USA hurricane observations)
  - 10 minutes (WMO, AUS, JAPAN +)
  - 2 minutes (USA)
  - 1 hour (UK)
- Why would averaging times differ?
  - Historic reasons
  - Prevalent storm types

Tropical Cyclone

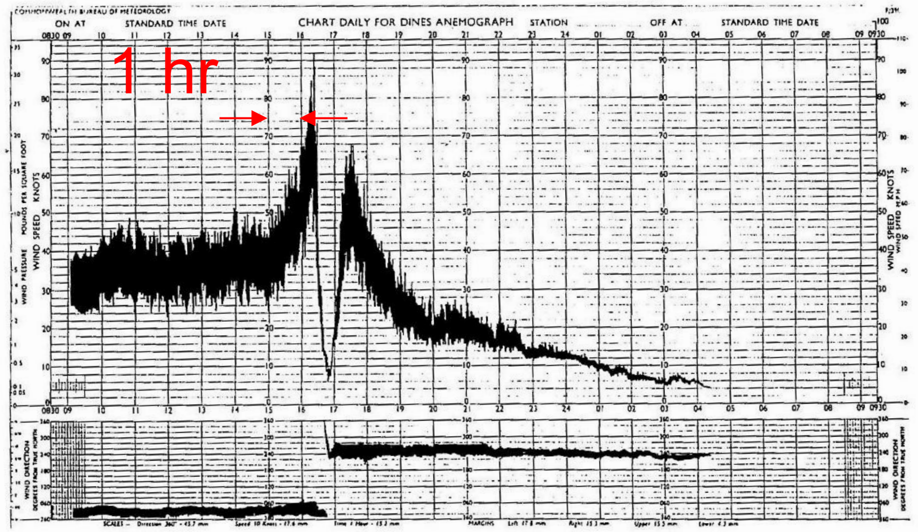
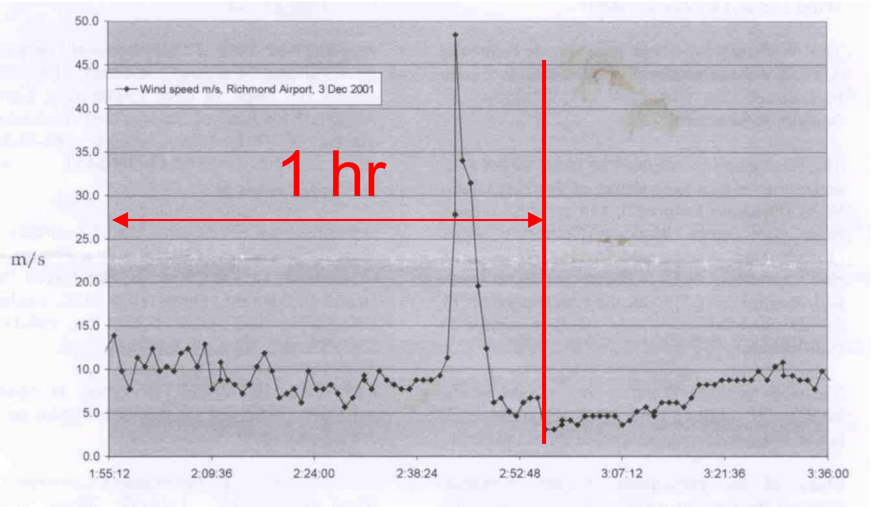
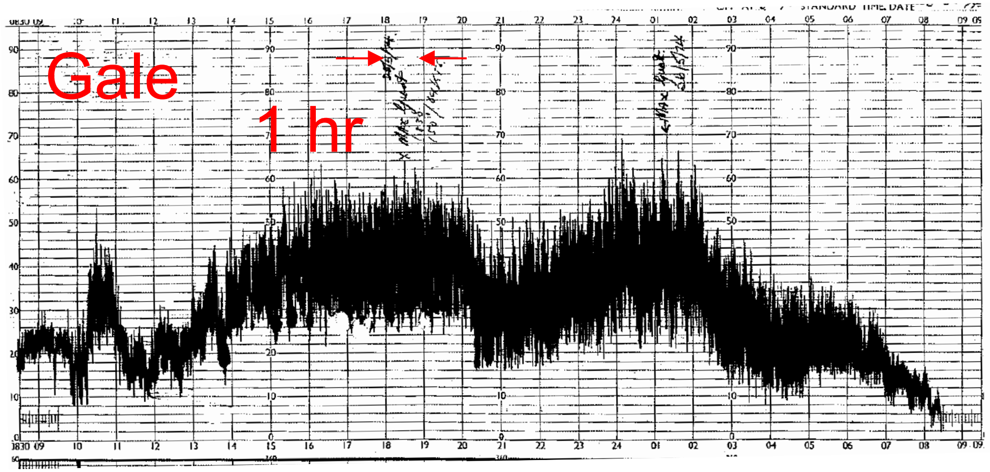


Figure 1.7 Anemograph from a tropical cyclone. Time unit: hours.

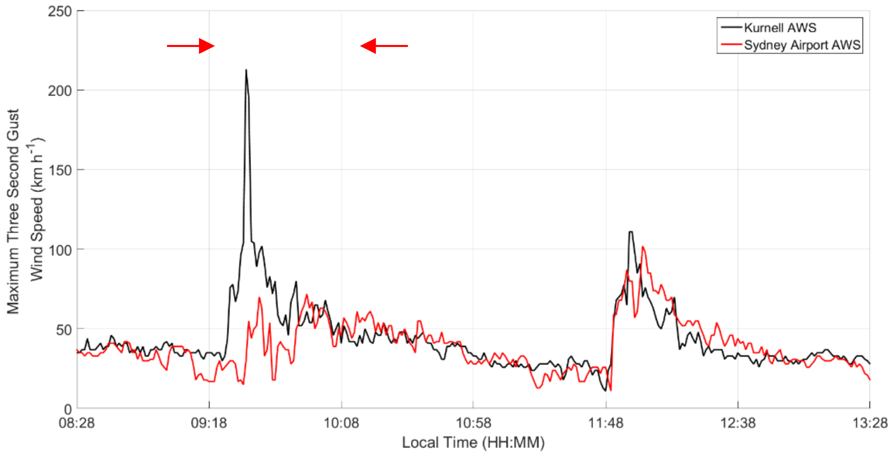
Thunderstorm



Gale



Tornado



Fluctuating wind

- Eddies/vortices on a wide range of sizes moving at the mean wind speed (Taylor hypothesis).
- Commonly represented by the standard deviation – *this has no 'physical' meaning.*

Variance

$$\sigma^2 = \frac{1}{T}\int_0^T \left(V(t) - \bar{U}\right)^2 dt$$

Standard deviation

$$\sigma = \sqrt{\frac{1}{T}\int_0^T \left(V(t) - \bar{U}\right)^2 dt}$$

σ_u, σ_v, σ_w,

Fluctuating wind

- Standard deviation is typically transformed into the so-called *turbulence intensity*, $I_{u,v,w}$.
- Mathematically, $I_{u,v,w}$, is the coefficient of variation.

$$I_u = \sigma_u / \bar{U} \text{ (longitudinal)}$$

$$I_v = \sigma_v / \bar{U} \text{ (lateral)}$$

$$I_w = \sigma_w / \bar{U} \text{ (vertical)}$$

*Note that all are divided by the mean along-wind (U) velocity

- In equilibrium flow,

Typical assumed value for σ_u (observations: 1.8 - 3.15)

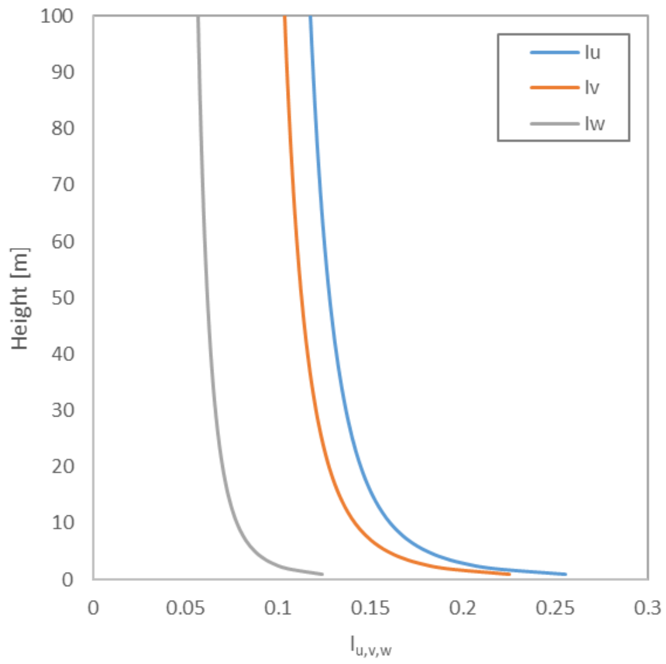
$$I_u = \frac{\sigma_u}{\bar{U}} = \frac{2.5u_*}{\left(\frac{u_*}{\kappa}\right) ln\left(\frac{z}{z_0}\right)} = \frac{1}{ln\left(\frac{z}{z_0}\right)}$$

Fluctuating wind

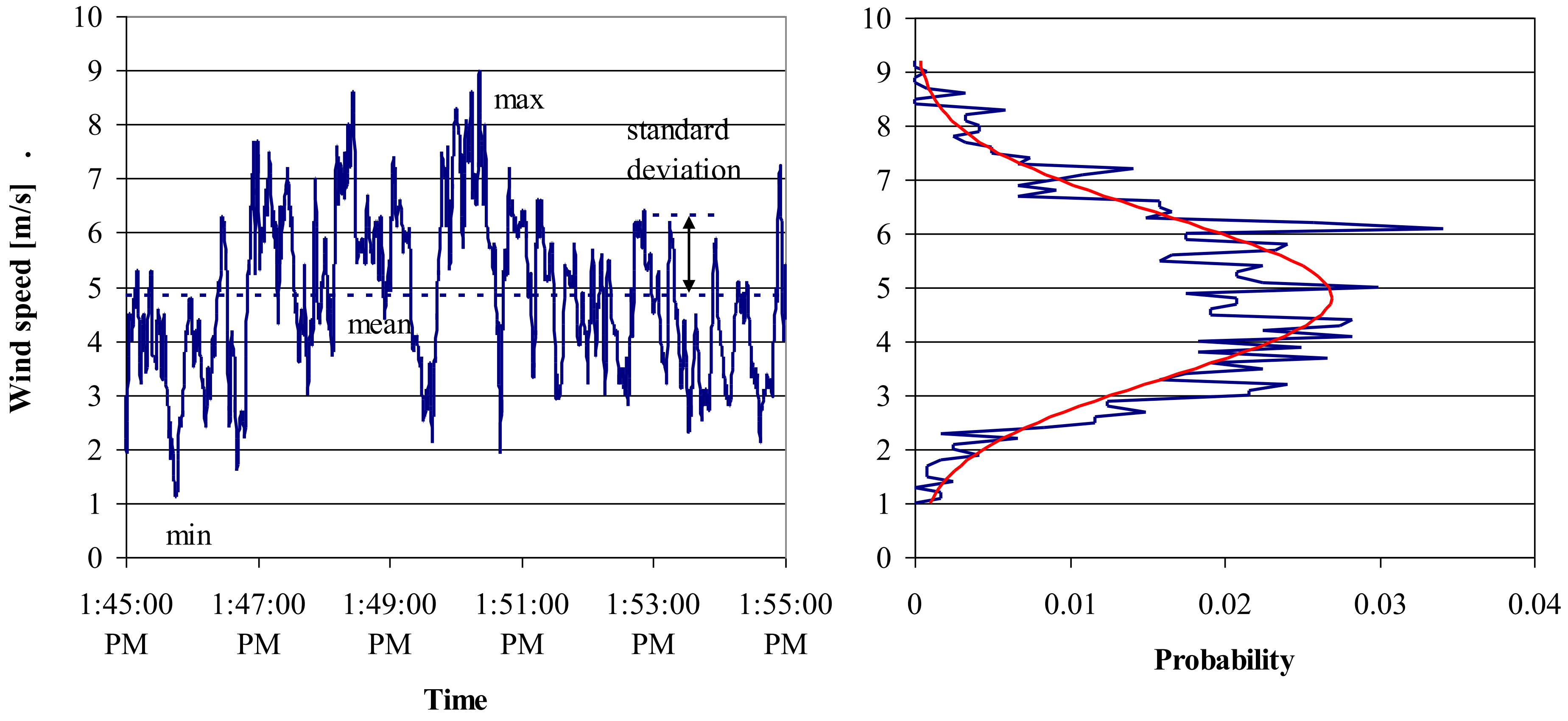
- Vertical and lateral turbulence is less than longitudinal turbulence.

$$I_v \cong \frac{0.88}{ln\left(\frac{z}{z_0}\right)} \qquad I_w \cong \frac{0.55}{ln\left(\frac{z}{z_0}\right)}$$

- How should $I_{u,v,w}$ change with height?



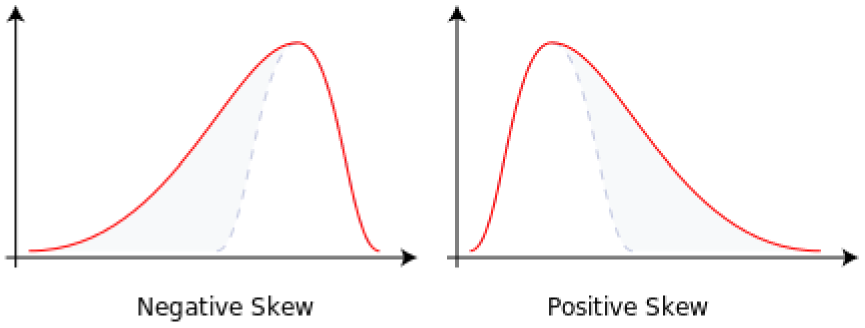
Probability density

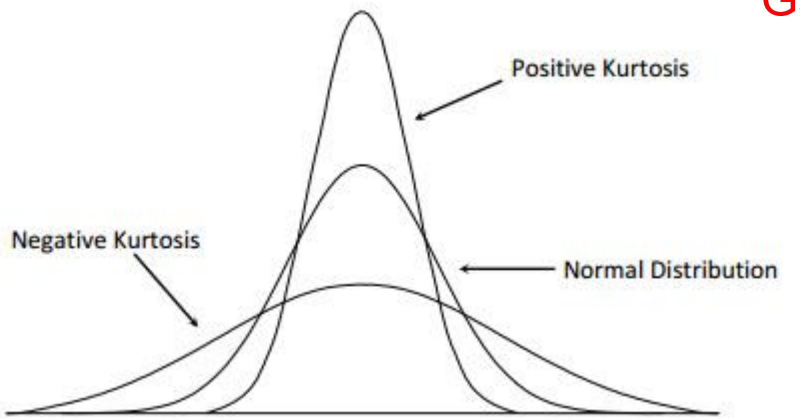


- Gaussian distribution (Normal) is generally assumed, and this is what most wind loading codes or standards are based upon.

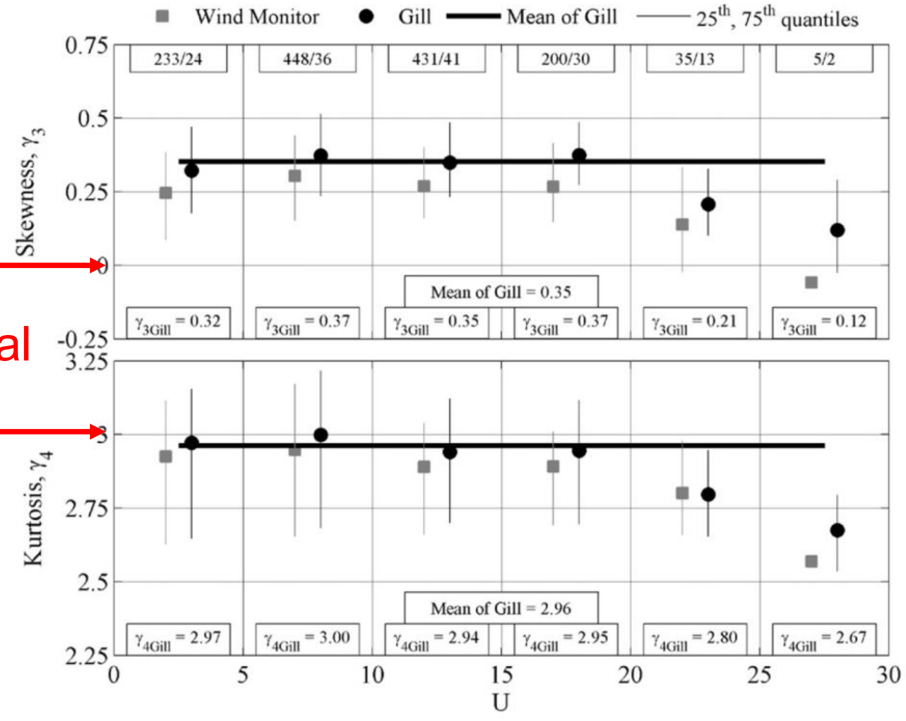
Probability density

- Research is showing the Gaussian assumption may not be valid near the surface (e.g. <10 m).
- This has implications for wind-resistant design codes (i.e. if the wind is skewed the loading will have greater skew).

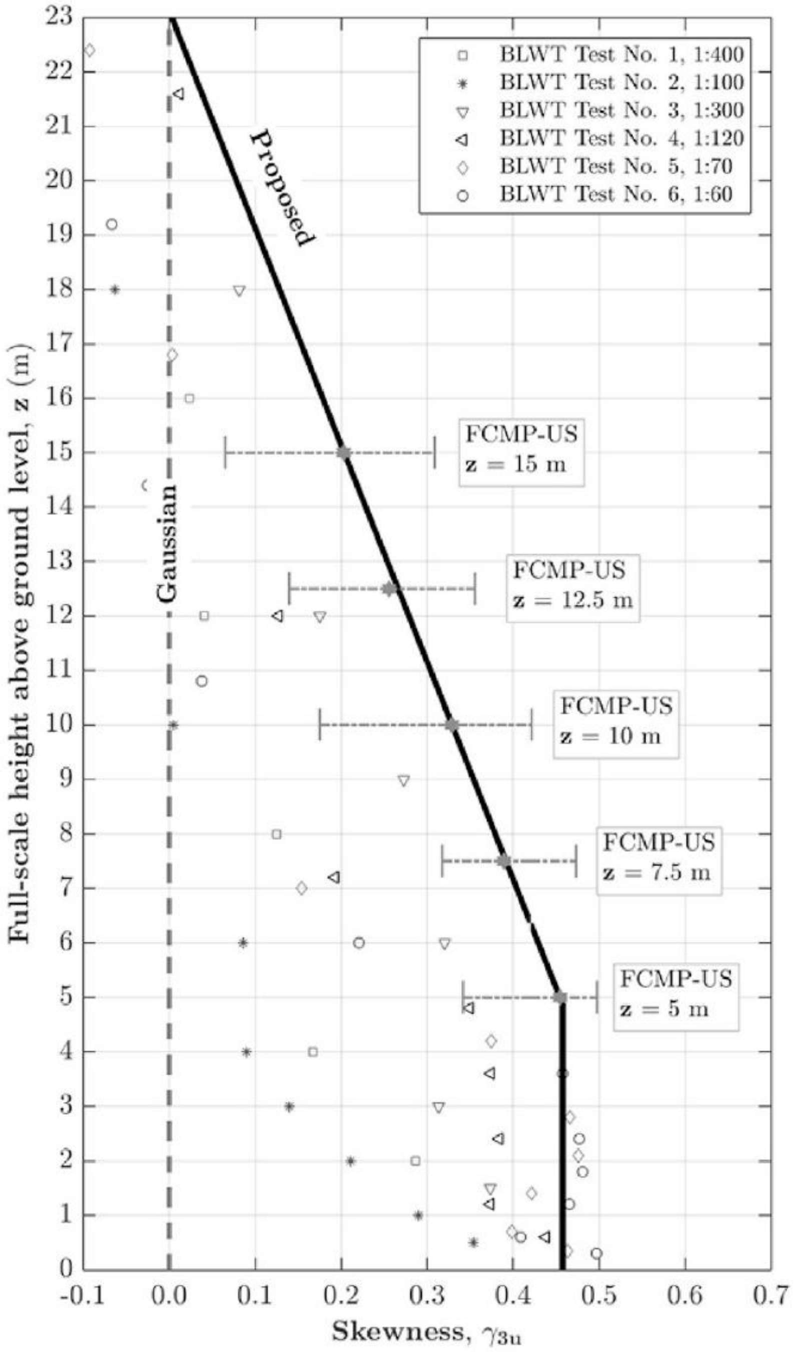




Gaussian/Normal



Balderrama et al. (2012)



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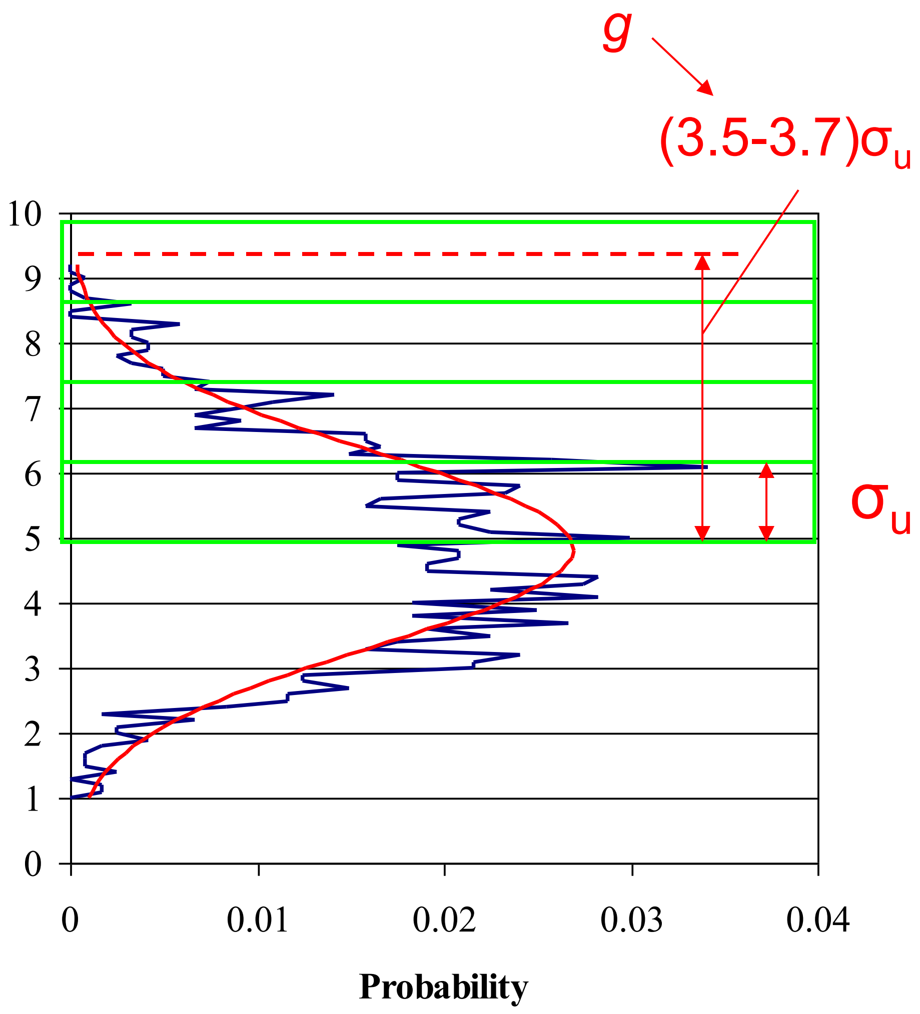
Fernandez-Caban & Masters, 2017

Peak factor model

$$\hat{U} = \overline{U} + g\sigma_u$$

Peak gust

Peak factor
f(averaging times)



The peak factor, *g*, describes how many standard deviations above your mean the peak gust is expected to sit.

Gust factor model

- A simpler relationship often used relates the peak gust to the mean.

$$G = \frac{\hat{U}}{\overline{U}}$$

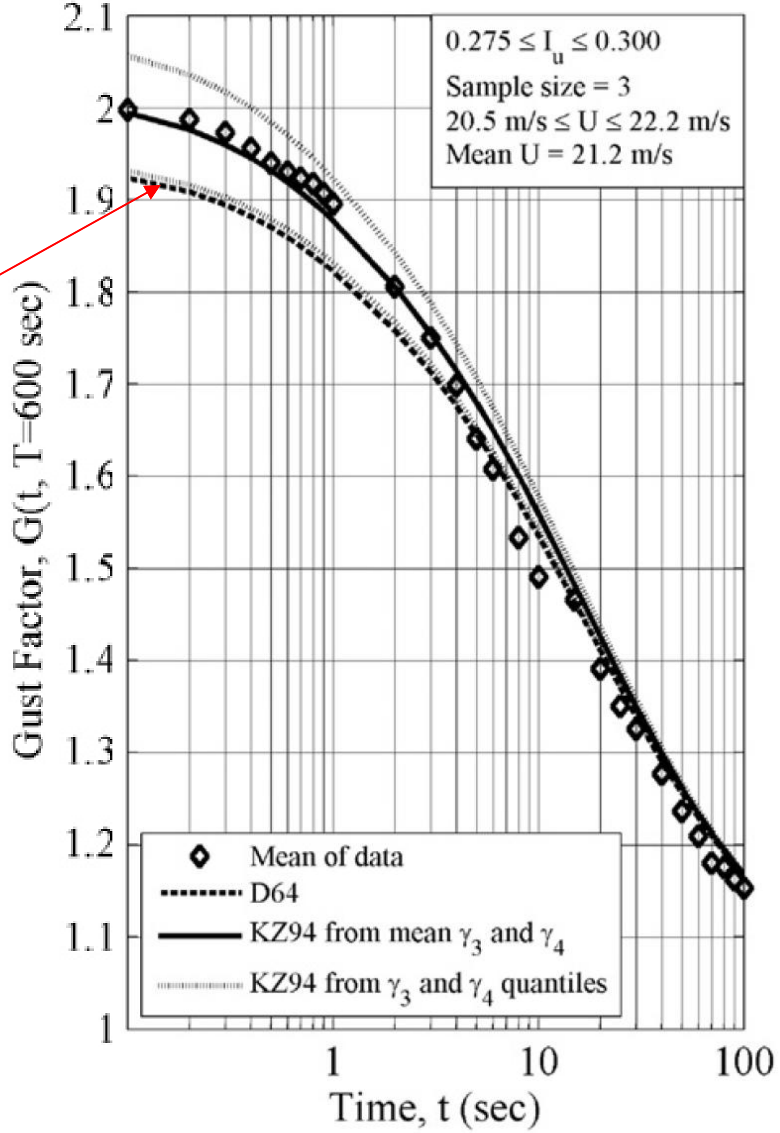
- This can empirically be approximated through

$$G = \frac{\hat{U}_{t,\mathrm{sec}}}{\overline{U}_{T,\mathrm{sec}}} = 1 + 0.5 I_u \ln\left(\frac{T}{t}\right)$$

- But more theoretically appealing solutions have been found.

Gust factors

Gaussian



Balderrama et al. (2012)

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Covariance & Correlation

- Gives information on the temporal and spatial distribution of the wind
  - Large structures - spatial correlation
  - Dynamically sensitive structures – temporal correlation
- Covariance – mean product of the fluctuating components at one or more points either simultaneously or with a time lag

Covariance of records *i* & *j* given time lag *τ*

Time lag

$$C_{ij}(\tau) = \overline{i(t)\cdot j(t+\tau)} = \lim T \to \infty \frac{1}{T}\int_0^T i(t)\cdot j(t+\tau)\,dt$$

Fluctuating component of record 1

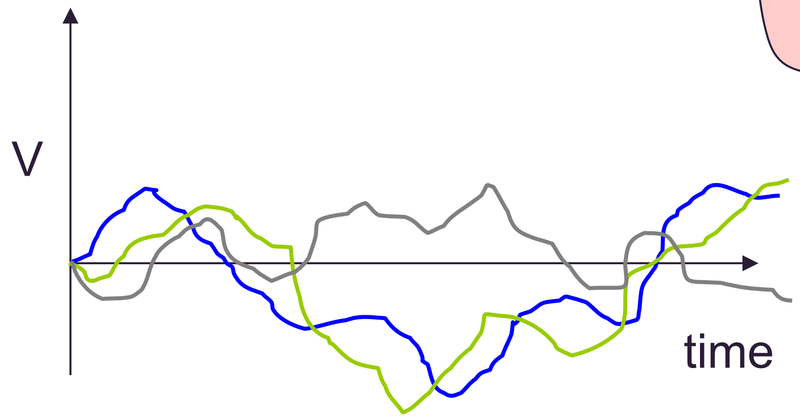
Fluctuating component of record 2

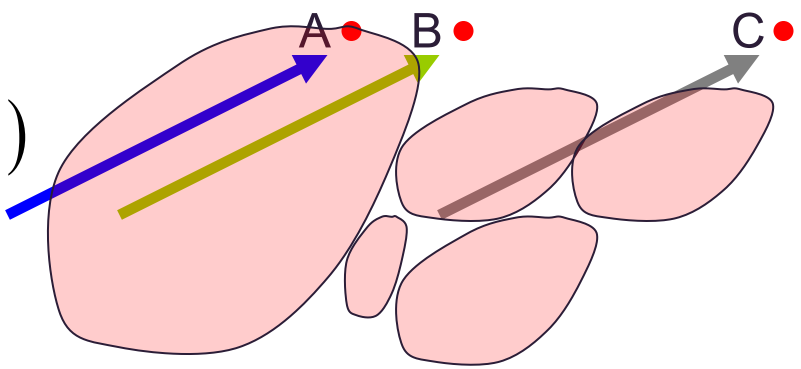
Covariance in space

$$C_{ij}(\tau) = \frac{1}{T}\int_0^T i(t)\cdot j(t+\cancelto{0}{\tau})dt$$

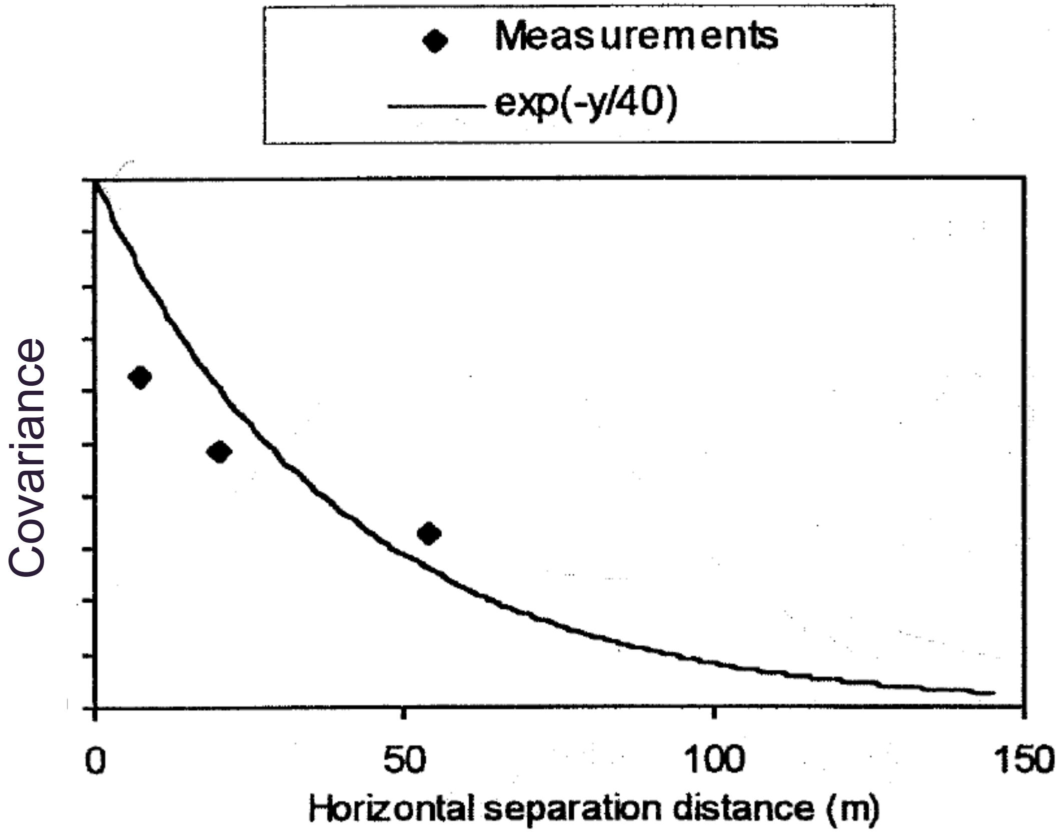
$$i(t) = V_A(z_A, t) - \overline{U}(z_A)$$

$$j(t+\tau) = V_B(z_B, t+\cancelto{0}{\tau}) - \overline{U}(z_B)$$





Covariance in space

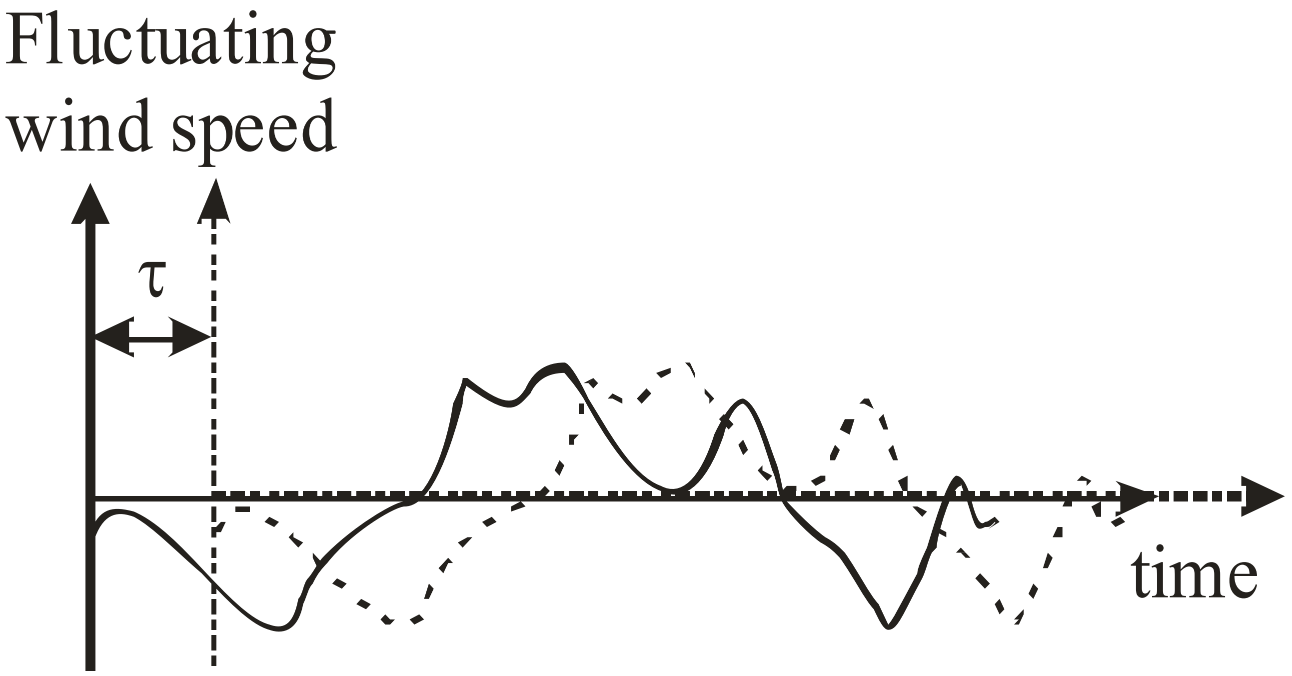


Covariance in time

$$C_{ij}(\tau) = \frac{1}{T}\int_0^T i(t)\cdot j(t+\tau)dt$$

$$i(t) = V_A(z_A, t) - \overline{U}(z_A)$$

$$j(t+\tau) = V_B(z_B, t+\tau) - \overline{U}(z_B)$$



Covariance

- Measurements at a single point – autocovariance.
  - Temporal information.
- Measurements at multiple points – covariance.
  - Temporal and spatial information.
- Nine possible covariance functions.
  - Important: autocovariance, uv, and uw.
- At $\tau = 0$, autocovariance = variance, therefore useful to normalise using variance.

Correlation

- Correlation coefficient (normalised covariance)

Correlation coefficient

$$\rho_{ij} = \frac{C_{ij}}{\sigma_i \cdot \sigma_j}$$

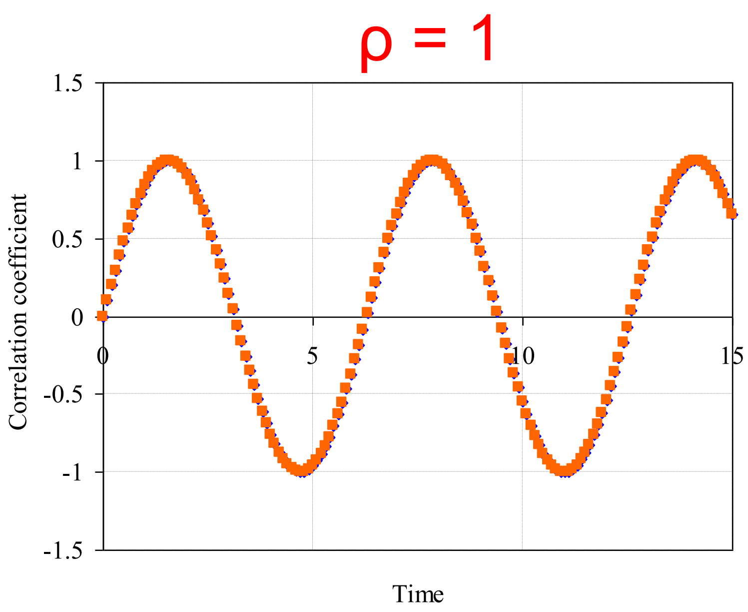
$-1 \leq \rho \leq 1$

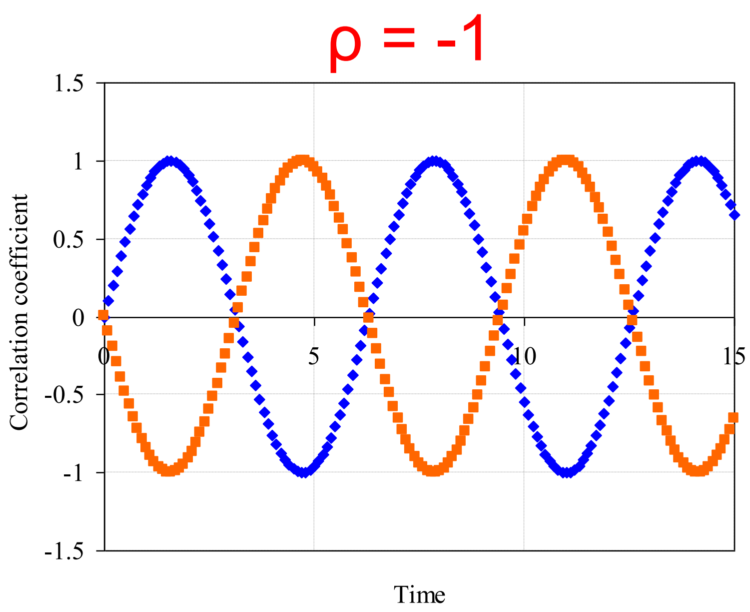
- Autocorrelation high when τ is small

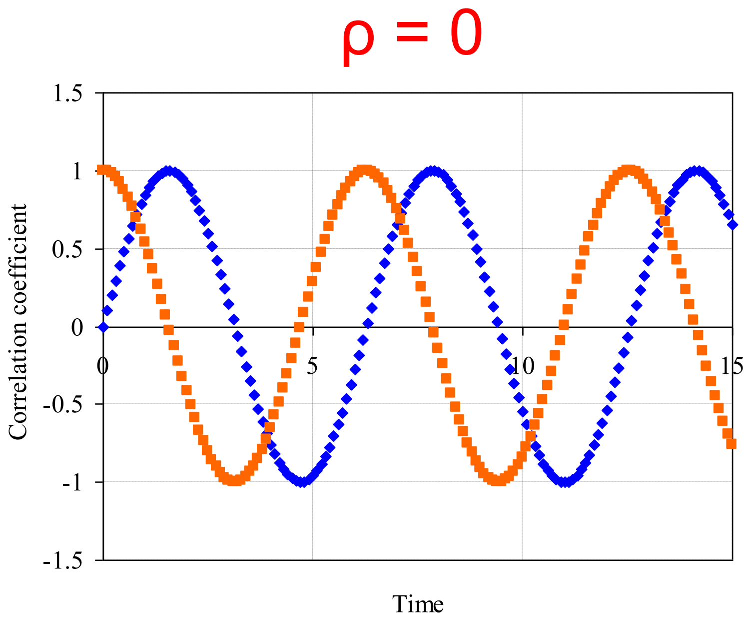
- As τ increases, autocorrelation decreases

- Cross-correlation high when spacing between measurement locations is small and decreases with spacing

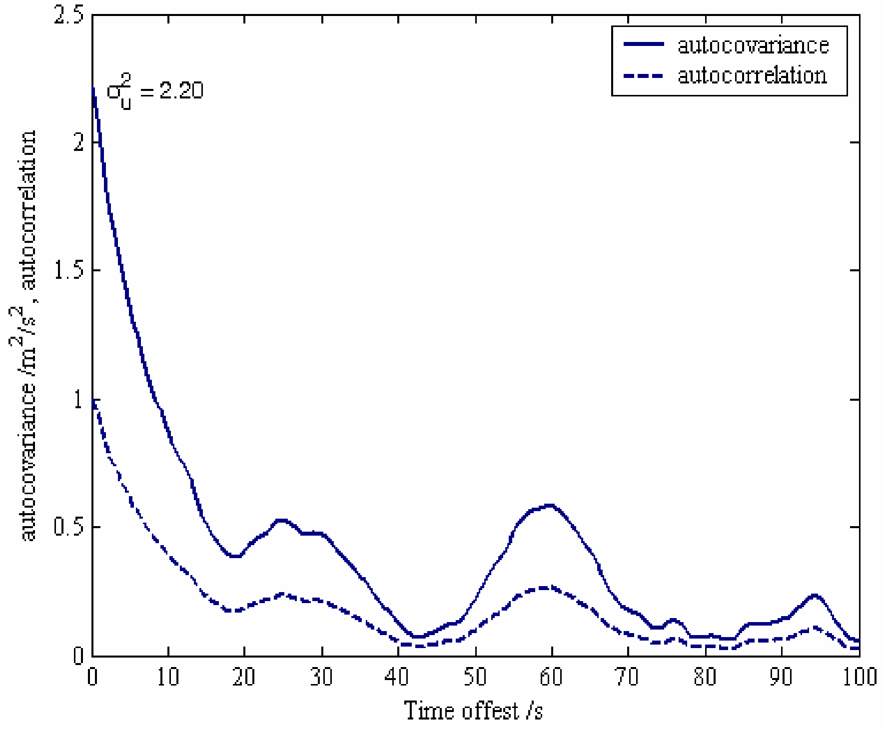
Correlation



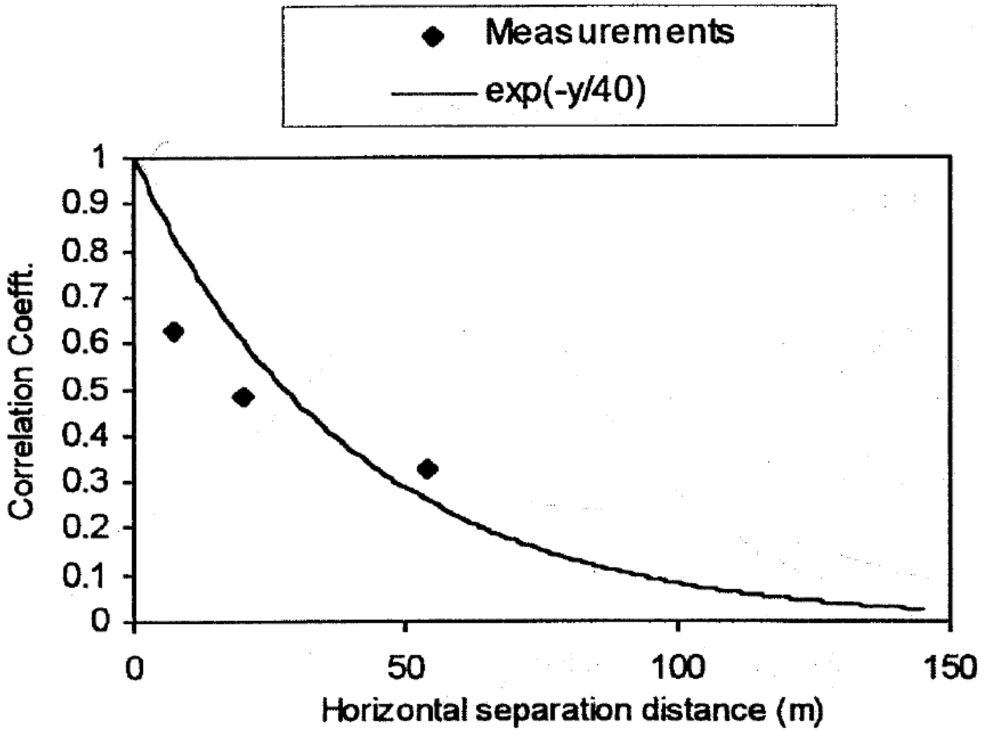




Example data

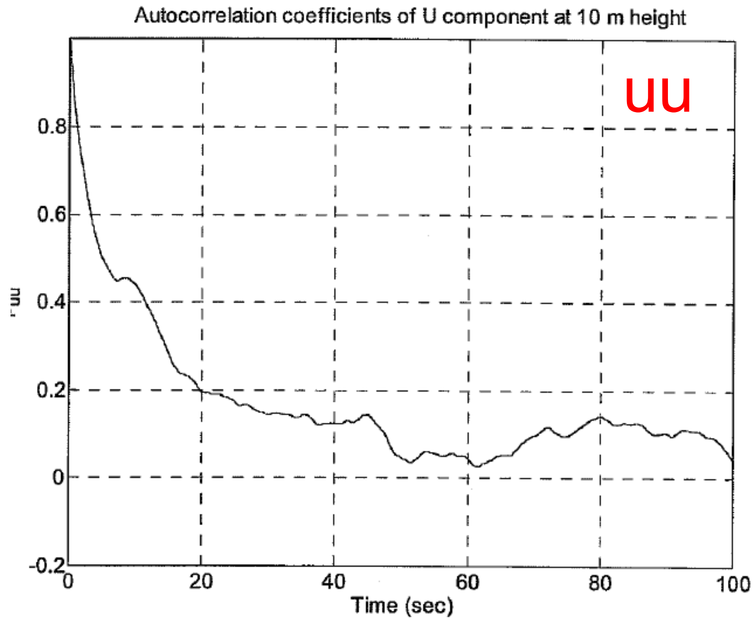


Temporal

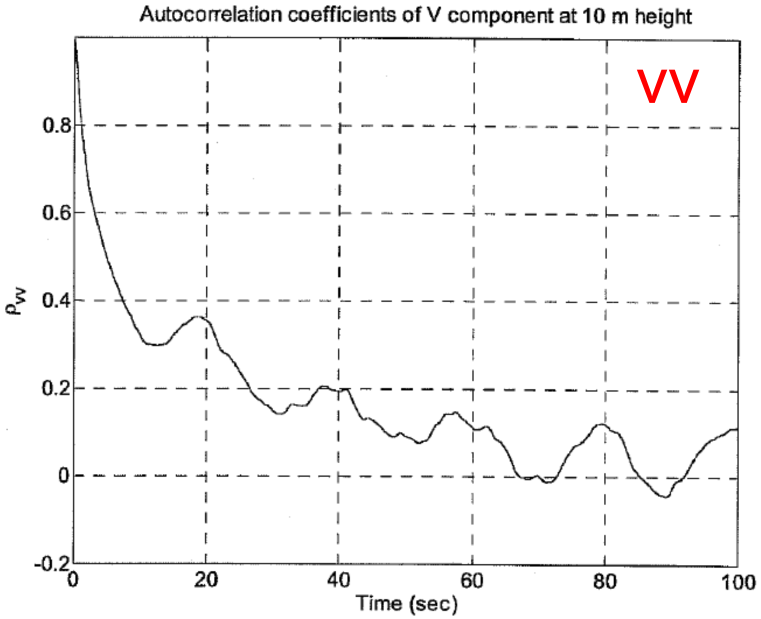


Spatial

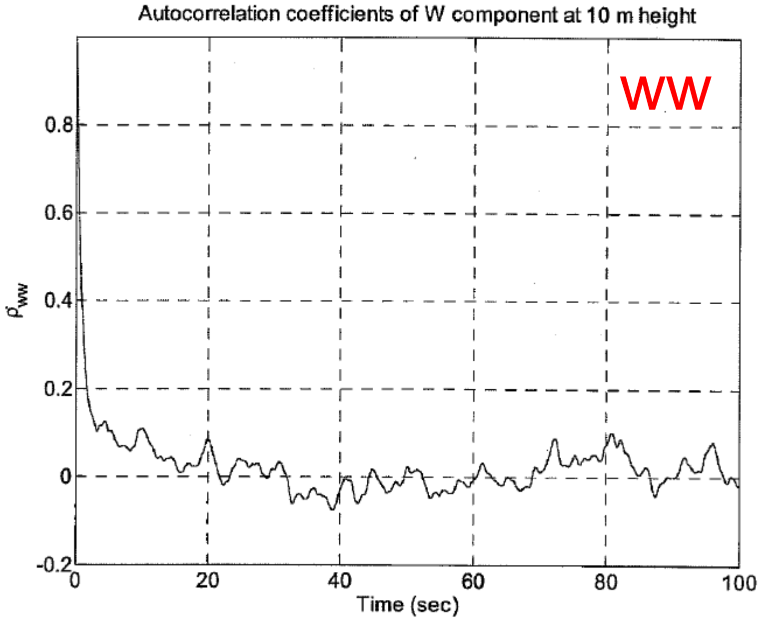
Example data



(a) U component



(b) V component



(c) W component

Gust size

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Using auto-correlation information we can define the duration and size of a *typical* gust

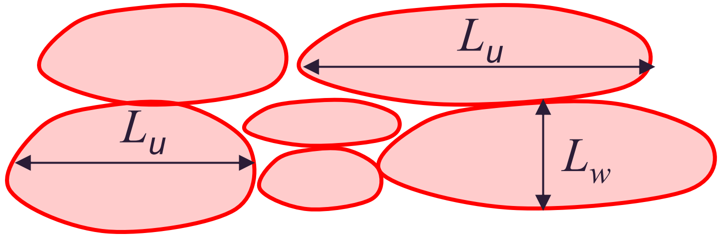
Time scale

ρ_{uu} , ρ_{vv}, ρ_{ww}

$$T_i = \int_0^\infty \rho_{ii}(\tau)\,d\tau$$

Length scale

$$L_i = T_i \overline{U}$$



Gust size

Length scale L_u

$$L_u = \int_{-\infty}^{\infty} \rho_{ij}(\tau = 0, s)\,ds$$

Cross correlation coefficient

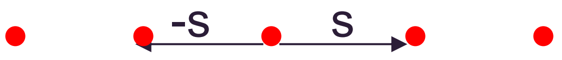


Table 1 Summary of 10 m height wind speed data in Fig. 2. Mean velocity = 12.3 m/s.

Component i	I_i	T_i (s)	L_i (m)
u	0.144	17.4	213
v	0.130	15.8	193
w	0.068	2.38	27.4

Turbulence quantities

- I_u, G, g describe the magnitude of turbulence.
- For dynamically sensitive structures the frequency of turbulence is important.
- Analysis moved from the time domain to the frequency domain.
- Spectral density (spectra, spectrum, power spectrum) of the wind describes the distribution of turbulence fluctuations with frequency, n.
- More specifically, the spectra describes the contribution to variance of winds in the range n to $n + dn$.

Contact

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